
Analog Electronic Circuits (EC2.103): Course Project

Spring 2026, IIIT Hyderabad, **Due date:** 05 May 2026

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Instructions:

1. Submit your project report in the given two-column format as a single PDF (Name1_Name2_Name3_GroupNumber.pdf) on Moodle by the due date. Only LaTeX, Word, or Notion submissions are allowed.
 2. The report should be self-explanatory and must include complete solutions. Answers must include Schematics, SPICE directives, *Annotated* waveforms, and Inference/Discussion of results.
 3. Clearly highlight the contributions of each group member during the presentation.
 4. The final Report submission will be post end semester exams and is a hard deadline. Final presentation, demonstration and Viva dates shall be announced on Moodle. Please plan your travel after 10th May 2026.
 5. Use the provided model files `TSMC_180nm.txt` (NMOS/PMOS) and `UA741.301` (op-amp) for all simulations.
 6. Specify MOSFET parameters W , L , A_S , A_D , P_S , and P_D in all simulations.
 7. Strictly, an OpAmp should not be used to amplify a signal, OpAmp-based buffers can be used if required.
 8. Any form of copying or cheating will result in an immediate F grade.
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Quadrature Down Converter:

Fig. 1 depicts a quadrature down converter (QDC) that is commonly used in modern-day wireless receivers (RX) such as Bluetooth, Wi-Fi and WLAN. Quadrature downconversion helps in interference mitigation and improves the quality of communication. In this project, we will implement a prototype of QDC for the given specifications.

As shown in Fig. 1, the input signal $v_{in} = A_1 \cos(\omega_{in} t)$ is mixed with $v_{OSC_I} = A_2 \cos(\omega_{OSC} t)$ and $v_{OSC_Q} = A_2 \sin(\omega_{OSC} t)$ to produce in-phase (v_{IF_I}) and quadrature-phase (v_{IF_Q}) intermediate frequency (IF) signals, respectively. The in-phase and quadrature-phase signals have a phase difference of 90° . Mixing of two signals is equivalent to their multiplication, as shown below.

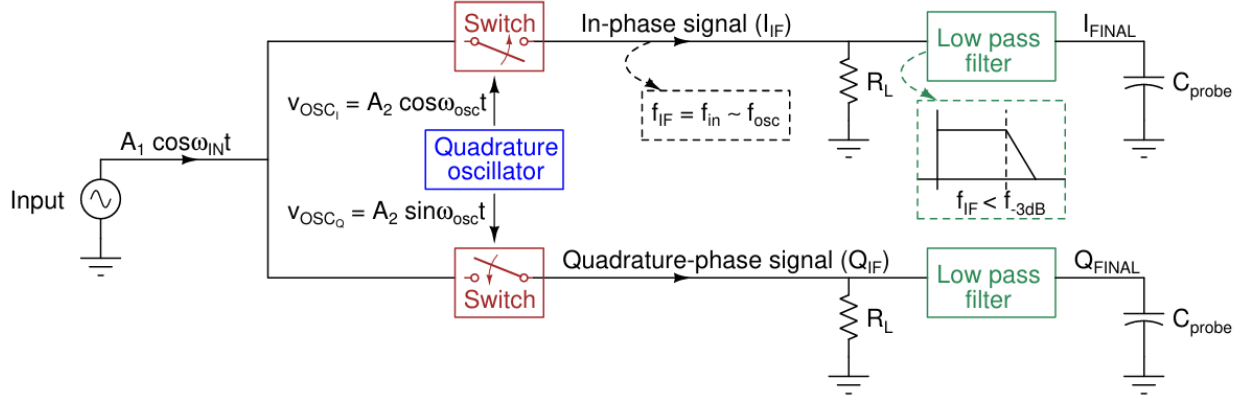


Figure 1: Block Diagram of QDC

$$v_{IF_I} = v_{in} \times v_{OSC_I} = \frac{A_1 A_2}{2} (\cos(\omega_{in} t - \omega_{OSC} t) + \cos(\omega_{in} t + \omega_{OSC} t))$$

$$v_{IF_Q} = v_{in} \times v_{OSC_Q} = \frac{A_1 A_2}{2} (\sin(\omega_{in} t + \omega_{OSC} t) - \sin(\omega_{in} t - \omega_{OSC} t))$$

As shown in Fig. 1, the mixed signal is fed to a low-pass filter to pass only the IF signal with frequency $\omega_{IF} = \omega_{IN} - \omega_{OSC}$, which can be made sufficiently low value for sufficiently high values of ω_{IN} and ω_{OSC} . For example, if $f_{IN} = 2.4$ GHz and $f_{OSC} = 2.401$ GHz, then $f_{IF} = 1$ MHz.

Q1. BONUS Question (not Mandatory to submit)

Briefly explain the utility/need of quadrature (I-Q) operation (down conversion/mixing) in modern-day wireless receivers. Support your answer with clear diagrams, spectrum depictions and mathematical expressions.

References

- [1] A. A. Abidi, "Direct-conversion radio transceivers for digital communications," *IEEE Journal of Solid-State Circuits*, vol. 30, no. 12, pp. 1399–1410, Dec. 1995.
- [2] B. Razavi, *RF Microelectronics*, 2nd ed. Upper Saddle River, NJ, USA: Prentice Hall, 2011, ch. 3–4.

Q2. Quadrature Oscillator Design

Using opamps, design a quadrature oscillator which produces two sinusoidal signals (v_{OSC_I} & v_{OSC_Q} at allocated frequency (150 KHz or 200 KHz) with a phase difference of 90° and oscillation amplitude of $1 V_{PP}$.

- a. Clearly show the topology, calculations for finding component (V_{DD} , V_{SS} , R , C (as applicable)) values and schematic with *annotated* component values.
- b. Use the given model file for the opamp for LTSpice simulations. Give plots for v_{OSCI} & v_{OSCQ} from the transient simulation and FFT. Clearly mark the phase difference between two signals on transient plots.
- c. Realize the circuit in the lab by using a 741 opamp IC and other passive components. From the measurements, report the transient waveforms and FFT spectrum for the two signals. Please note that you may need to tweak some parameters in the actual experiment compared to the simulation setup. Report the changes made in the values.

References

- [1] A. S. Sedra and K. C. Smith, *Microelectronic Circuits*, 7th ed. New York, NY, USA: Oxford University Press, 2015, ch. 2, ch. 14.
- [2] R. Mancini, "Design of op amp sine wave oscillators," Texas Instruments, 2000.
- [3] R. Holzel, "A simple wide-band sine wave quadrature oscillator," *IEEE Transactions on Instrumentation and Measurement*, vol. 42, no. 3, pp. 758–760, Jun. 1993.

Q3. Switch (mixer) design

As shown in Fig. 2, a simple MOSFET can be used as a Switch (mixer), where the oscillator signal is applied to the gate of the MOSFET, the Input is applied at the source and the Intermediate frequency output is taken at the drain end.

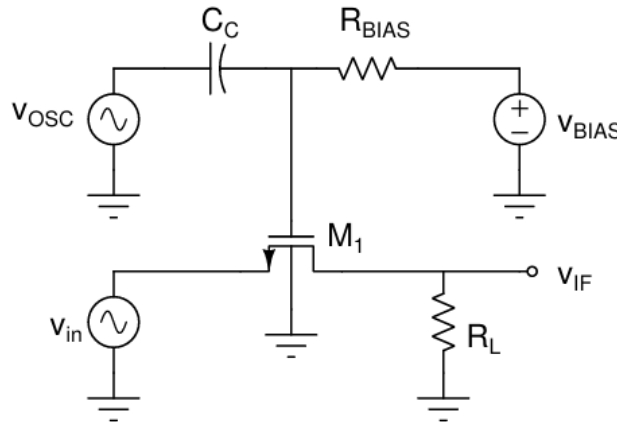


Figure 2: NMOS Mixer Circuit

- a. Detail the specific MOSFET dimensions and parasitic parameters (W , L , A_S , A_D , P_S , and P_D) utilized in the LTSpice simulations. Please note that while physical lab MOSFET have fixed sizing, these parameters should be optimized within the simulation environment.

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- b. As shown in Fig. 2, implement the mixer using an NMOS in LTSpice. Clearly mention the values of R_{BIAS} , C_c .
 - c. Consider $R_L = 1k\Omega$, the amplitude of v_{in} is $100mV$, and the frequency is f_{IN} .
 - d. i. For 150KHz frequency, Show transient plots of v_{in} , v_{IF} for $f_{IN} = 145$ kHz, 148 kHz, 149 kHz, 151 kHz, 152 kHz, 155 kHz . Also show the corresponding FFT plots for v_{IF} .
[OR]
 - ii. For 200KHz frequency, Show transient plots of v_{in} , v_{IF} for $f_{IN} = 195$ kHz, 198 kHz, 199 kHz, 201 kHz, 202 kHz, 205 kHz . Also show the corresponding FFT plots for v_{IF} .
 - e. Realise the circuit in lab using the NMOS transistors and report above component values and plots from measurements. Please note that the NMOS in lab might differ from the one that you use in simulations. For giving v_{in} and v_{osc} to the mixer, you can use two function generators in the lab.

Q4. Low pass filter design

- a. Design a low-pass RC filter (LPF) with a $-3dB$ cut-off frequency of 3 kHz. Show the component values with calculations.
- b. Report frequency response (output magnitude vs frequency) from AC analysis in LTSpice simulations.
- c. Report transient response for 1 kHz signal and 10 kHz using LTSpice simulations.
- d. Realise the circuit in the lab and report AC and transient plots from measurements.
- e. Connect mixer setup in the previous part and LPF and report results from simulations and measurements.
- f. Design an Amplifier stage to get the final output Voltage with an allocated amplitude of 200mV (i.e, 400mV peak to peak) or 250mV (i.e, 500mV peak to peak).

Q5. Complete circuit prototype design

Connect all building blocks (oscillator, mixer, filter) and make the complete circuit shown in Fig. 1. Clearly show and tabulate all component/supply values in the schematic.

- a. Run transient simulations and clearly show waveforms (input, oscillator, IF, IF (FINAL)). Clearly annotate to show the phase difference between the I-Q components.
- b. Report FFT plots of IF (FINAL- I and Q both). Can you process the FFT plots to find the phase of the final I and Q components?
- c. Realise the complete circuit in the lab and report transient plots of v_{in} , v_{OSC_I} , v_{OSC_Q} , v_{IF_I} , v_{IF_Q} , $v_{IF_{FINAL_I}}$ and $v_{IF_{FINAL_Q}}$ from measurements. Clearly show the corresponding phase difference in the I-Q plots for every result.

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- d. Report FFT plots of $v_{IF_{FINAL_I}}$ and $v_{IF_{FINAL_Q}}$ from measurements.
 - e. Make a comparison table for simulation and measurement results showing frequencies (I/Q), phase difference (I/Q), amplitude (I/Q), supply-voltages, bias-voltages, etc., at each relevant node. Also, compare the component values used in the simulation and measurements.

Parameters	Simulated	Measured
Oscillator Frequency		
Oscillator Amplitude (I-phase)		
Oscillator Amplitude (Q-phase)		
Input frequency		
IF		
Supply		
V_{BIAS}		
C_C		
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Table 1: Performance summary and comparison

Q6. Project report

Make a comprehensive project report in IEEE two-column format (shared along with the project problem).

Other references

- [1] Latex template can be found at the following link: [IEEE Conference Template](#)
- [2] Lecture notes, tutorials, and labs conducted in this course